Realistic Bodycam FPS System

Important notice about the weapon models

This update includes **simple temporary blockout weapon models**, allowing you to immediately test the shooting, aiming, animation, reloading, flashlight, and weapon-switching systems. Please note that these are placeholder models and are not the final visual assets.

The original, fully detailed weapon and arm models are currently being developed and will be included in the next update. I sincerely apologize that the final models are not available yet and for any inconvenience this may cause.

Thank you for your patience and support.

About this update

This update significantly expands the original asset with a complete modular weapon workflow, improved bodycam movement, shooting and reloading features, interactive training targets, player health, and a diegetic death system.

Every major feature can be enabled, disabled, and adjusted directly in the Inspector.

Temporary blockout weapons are included and already configured to demonstrate the new systems. They can be replaced with your own models at any time.

Main improvements

Bodycam controller

- Improved camera movement, sway, roll, inertia, and body weight.
- More responsive and natural mouse movement.
- Walk, run, head bob, step rotation, and idle breathing.
- Free Look mode using `Left Alt` by default.
- Weapon compensation during Free Look.
- Multiple customizable bodycam presets.
- Randomize Bodycam Feel option.
- Modular Inspector toggles for individual features.

Weapon movement and aiming

- The weapon reacts to mouse movement before the camera finishes following it.
- Faster independent weapon rotation response.
- Positional and rotational weapon lag.
- Movement inertia, idle motion, walking bob, and running bob.
- Adjustable aiming position, rotation, speed, and FOV.
- Reduced movement while aiming.
- New aim-release wobble: the weapon hesitates and oscillates slightly when returning from aim instead of moving back in a perfectly straight line.

Weapon switching

- Simple modular weapon switching system.
- Supports the mouse wheel and number keys `1–9`.
- Adjustable switch delay.
- Diegetic switch animation: the current weapon moves down and out of the camera before the next weapon appears.

Advanced shooting system

- Raycast-based shooting.
- Optional physical projectiles.
- Configurable bullet prefab, spawn point, speed, lifetime, gravity, and collision behavior.
- Semi-automatic and automatic fire support.
- Fire rate and range controls.
- Configurable hit layers.
- Weapon recoil and camera recoil.
- Randomized recoil for less repetitive shooting.
- Muzzle flash particle support.
- Temporary muzzle Point Light when firing.
- Shooting audio with configurable clips.
- Animation support for idle, firing, and reloading.

Ammunition and reloading

- Optional ammunition system.
- Full Magazine reload mode for pistols, rifles, and similar weapons.
- Shell-by-Shell reload mode for shotguns.
- Adjustable magazine size and reload duration.
- Individual shell insertion timing.
- Continuous shotgun shell animation without returning to Idle between every shell.
- Optional shell reload interruption.
- Optional automatic reload when empty.
- Reload audio support.
- Configurable camera shake while reloading.
- Inspector fields are organized according to the selected reload type.

> `Shell Insert Duration` and the shell-specific settings are only required when using `Shell-by-Shell` reload mode. They are not required for `Full Magazine` weapons.

Flashlights

- Standalone realistic flashlight system.
- Separate weapon-mounted flashlight system.
- Simple on/off input using `F` by default.
- Independent activation and deactivation sounds.
- Weapon flashlight can be configured separately from a player/head flashlight.

Player health and diegetic death

- Configurable maximum health.
- Public methods for taking damage, healing, killing, and respawning.
- Optional invincibility.
- Events for damage, death, and respawn.
- Player controls and weapon systems can be disabled automatically on death.
- Weapon object can be hidden during death.
- Randomized fall direction.
- Loss-of-balance animation before falling.
- Strong, irregular bodycam sway during the fall.
- Ground impact, bounce, shake, and settling motion.
- Continuous subtle camera movement while the player remains on the ground.
- Automatic blackout shortly after the impact.
- Optional death and ground-impact sounds.
- Optional automatic respawn.

Camera connection messages

- `CAM CONNECTED` message when the game starts.
- `CAM DISCONNECTED` message after death.
- `CAM CONNECTED` appears again after respawning.
- Separate colors and sounds for connection and disconnection.
- Separate timing settings for the connected and disconnected messages.
- Typing effect, low-FPS flicker, alpha fade, and position jitter.
- Optional message background.
- Respawn automatically waits for `CAM DISCONNECTED` to finish.

Training range systems

- Pop-up targets activated by trigger areas.
- Targets can rise from different starting rotations.
- Shootable targets return to their lowered position when hit.
- Static targets can fall when hit and reset repeatedly.
- Infinite target respawn support.
- Projectile and raycast hit compatibility.
- Optional impact prefab support.
- Low-FPS target detection boxes that dynamically fit visible targets.
- Configurable target markers and labels.

Additional immersion systems

- Dynamic walking and running footsteps with audio variation.
- Camera movement and equipment/cloth audio.
- Stamina breathing audio.
- Running wind audio.
- One-shot collision sounds.
- Player interaction with rigidbodies.
- Automatic door closing support.
- Animated bodycam overlay motion.

Default controls

Action	Default control
Move	`WASD`
Look	Mouse
Run	`Left Shift`
Jump	`Space`
Free Look	`Left Alt`

Fire	`Left Mouse Button`
Aim	`Right Mouse Button`
Reload	`R`
Switch weapon	Mouse wheel or `1-9`
Flashlight	`F`

Controls exposed in the Inspector can be changed per component.

Installation

1. Import the package into a Unity URP project.
2. Open the included Demo Scene or drag the player prefab into your scene.
3. Make sure the player references are assigned in the Inspector.
4. Use the included temporary blockout weapons or replace them with your own models under the weapon holder/follow target.
5. Configure its aim position, shooting system, animations, muzzle flash, and audio.
6. Confirm that your project Input Handling is configured correctly.

Important Input Handling configuration

The current controller uses Unity's classic `Input` API. In newer Unity projects, open:

`Edit > Project Settings > Player > Other Settings > Active Input Handling`

Select one of these options:

- `Both` — recommended.
- `Input Manager (Old)` — also supported.

If Unity asks to restart the Editor, accept the restart. If the project is

configured as `Input System Package (New)` only, the default player controls may not respond.

Adding your own weapon

1. Parent your weapon/arms object to the weapon holder or weapon follow target.
2. Reset or adjust its local position and rotation.
3. Add `BodycamWeaponMotionWithLag` to the appropriate weapon transform.
4. Add `BodycamAdvancedShootingSystem` to the weapon.
5. Assign the player camera, muzzle point, muzzle particle, muzzle light, audio source, and optional projectile prefab.
6. Assign the Animator and configure the Idle, Fire, and Reload state names.
7. Add the weapon to `BodycamWeaponSwitcher` if you are using multiple weapons.

Full Magazine and Shell-by-Shell reloads

Use **Full Magazine** for weapons that replace or refill the complete magazine in one animation.

Use **Shell-by-Shell** for shotguns or weapons that insert ammunition individually. Configure the shell animation and insertion duration only for this mode.

Physical projectile setup

For a physical bullet:

1. Create a projectile prefab.
2. Add a Collider.

3. Add a Rigidbody.
4. Add `BodycamProjectile`.
5. Assign an optional impact prefab.
6. Assign the projectile prefab and spawn point in the shooting system.

The Impact Prefab is the visual effect spawned when the projectile hits something. It can be left empty if no impact effect is needed.

Health and damage setup

1. Add `BodycamPlayerHealth` to the main player object.
2. Assign the camera transform.
3. Add the movement, shooting, weapon motion, and weapon switching components to `Systems To Disable`.
4. Assign the weapon root.
5. Assign `BodycamStartupMessage` to `Camera Status Message`.
6. Enable `Auto Respawn` if required.
7. Add `BodycamDamageSource` to projectiles, enemies, or trigger volumes that should damage the player.

Damage can also be applied through code:

```
playerHealth.TakeDamage(25f);
```

Compatibility

- Unity 2021 or newer.
- Universal Render Pipeline (URP).
- Classic Input Manager or Active Input Handling set to Both.

Support

If something does not work as expected, please include the following information when contacting support:

- Unity version.
- Render pipeline and version.
- Active Input Handling setting.
- Screenshot of the Console errors.
- Screenshot of the relevant Inspector component.
- Clear description of the steps needed to reproduce the issue.

Thank you for using Realistic Bodycam FPS System.